Multiple Choice:

1. Magnetic flux can always be attributed to:

   A. Ferromagnetic materials
   B. Aligned atoms
   C. Motion of charged particles
   D. The geomagnetic field

2. A straight conductor 500 mm long is moved with constant velocity at right angles both to its length and to a uniform magnetic field. Given that the e.m.f. induced in the conductor is 2.5 V and the velocity is 5 m/s, calculate the flux density of the magnetic field. If the conductor forms part of a closed circuit of total resistance 5 ohms, calculate the force on the conductor.

   A. 0.25 N C. 2.5 N
   B. 0.025 N D. 250 N

3. Which of the following statements is true?

   A. Unlike charges repel each other.
   B. Like charges repel each other.
   C. Unlike charges attract each other.
   D. Both B and C.

4. A changing magnetic field:

   A. Produces an electric current in an insulator.
   B. Magnetizes the earth.
   C. Produces a fluctuating electric field.
   D. Results from a steady electric current.

5. A conductor of length 15 cm is moved at 750 mm/s at right-angles to a uniform flux density of 1.2 T. Determine the e.m.f. induced in the conductor.

   A. 0.135 V C. 0.0135 V
   B. 1.350 V D. 13.50 V

6. For the current carrying conductor lying in the magnetic field shown in the figure below, the direction of the force on the conductor is:

   A. downwards C. upwards
   B. to the right D. to the left

7. Which is the wrong statement? The magnetizing force at the center of a circular coil varies

   A. directly as the number of its turns
   B. directly as the current
   C. directly as its radius
   D. inversely as its radius

8. A 25 cm long conductor moves at a uniform speed of 8 m/s through a uniform magnetic field of flux density 1.2 T. Determine the current flowing in the conductor when its ends are connected to a load of 15 ohms resistance.

   A. 1.6 A C. 16 A
   B. 0.16 A D. 0.016 A

9. Two bar magnets are placed parallel to each other and about 2 cm apart, such that the south pole of one magnet is adjacent to the north pole of the other. With this arrangement, the magnets will:

   A. attract each other
   B. have no effect on each other
   C. repel each other
   D. lose their magnetism

10. A car is travelling at 80 km/h. Assuming the back axle of the car is 1.76 m in length and the vertical component of the earth's magnetic field is 40 μT, find the e.m.f. generated in the axle due to motion.

   A. 1.56 mV C. 0.156 mV
   B. 15.6 mV D. 156 mV

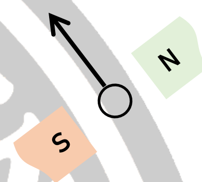
11. Calculate the current required in a 240 mm length of conductor of a d.c. motor when the conductor is situated at right-angles to the magnetic field of flux density 1.25 T, if a force of 1.20 N is to be exerted on the conductor.

   A. 4 A C. 2 A
   B. 6 A D. 5 A

12. If a neutral atom loses one of its valence electrons, it becomes a (n)

   A. negative ion.
   B. electrically charged atom.
   C. positive ion.
   D. Both B and C.

13. For the current-carrying conductor lying in the magnetic field shown in the figure below, the direction of the current in the conductor is:



   A. towards the viewer
   B. away from the viewer
   C. upwards
   D. downwards

14. A conductor 300 mm long carries a current of 13 A and is at right-angles to a magnetic field between two circular pole faces, each of diameter 80 mm. If the total flux between the pole faces is 0.75 mWb, calculate the force exerted on the conductor.

   A. 0.0582 N C. 0.582 N
   B. 5.82 N D. 58.2 N

15. Calculate the force exerted on a charge of $2x10^{-18}$ C travelling at $2x10^{6}$ m/s perpendicular to a field of density $2x10^{-7}$ T.

   A. 8.0×10^{-19} N
   B. 0.8×10^{-19} N
   C. 0.008×10^{-19} N
   D. 80.0×10^{-19} N

16. The most basic particle of negative charge is the ____.

   A. coulomb C. electron
   B. proton D. neutron

17. Which of the following statements is true?

   A. Unlike charges repel each other.

B. Like charges repel each other.
C. Unlike charges attract each other.
D. Both B and C.

18. Aluminum, with an atomic number of 13, has

A. 13 valence electrons.
B. 3 valence electrons.
C. 13 protons in its nucleus.
D. Both B and C.

19. The nucleus of an atom is made up of

A. electrons and neutrons.
B. ions.
C. neutrons and protons.
D. electrons only.

20. One ampere of current corresponds to

A. 1 C/1 s
B. 1 J/1 C
C. 6.25×10^{18} electrons
D. 0.16×10^{-18} C/s

21. Conventional current is considered

A. the motion of negative charges in the opposition direction of electron flow.
B. the motion of positive charges in the same direction as electron flow.
C. the motion of positive charges in the opposition direction of electron flow.
D. None of the above.

22. The coulomb is a unit of

A. electric charge.
B. potential difference.
C. current.
D. voltage.

23. A conductor 30 cm long is situated at right-angles to a magnetic field. Calculate the flux density of the magnetic field if a current of 15 A in the conductor produces a force on it of 3.6 N

A. 0.08 T
B. 0.008 T
C. 8.0 T
D. 0.80 T

24. The coulomb is a unit of:

A. Power
B. Voltage
C. Energy
D. Quantity of electricity

25. How long must a current of 100 mA flow so as to transfer a charge of 80 C?

A. 800 ms
B. 800 s
C. 800 μs
D. 80 s

26. If a neutral atom loses one of its valence electrons, it becomes a (n)

A. negative ion.
B. electrically charged atom.
C. positive ion.
D. Both B and C.

27. A current of 6 A flows for 10 minutes. What charge is transferred?

A. 360 C
B. 3600 mC
C. 36 C
D. 3600 C

28. Voltage drop is the:

A. maximum potential
B. difference in potential between 2 points
C. voltage produced by a source
D. voltage at the end of the circuit

29. Determine the speed of a 10^{-19} charge travelling perpendicular to a field of flux density 10^{-7} T, if the force on the charge is 10^{-20} N.

A. 10^7 m/s
B. 10^5 m/s
C. 10^6 m/s
D. 10^3 m/s

30. Which of the following statements is false?

A. For non-magnetic materials reluctance is high
B. Energy loss due to hysteresis is greater for harder magnetic materials
C. The remanence of a ferrous material is measured in ampere/meter
D. Absolute permeability is measured in henrys per meter

31. A conductor carries a current of 70 A at right-angles to a magnetic field having a flux density of 1.5 T. If the length of the conductor in the field is 200 mm, calculate the force acting on the conductor.

A. 21.0 N
B. 0.21 N
C. 2.10 N
D. 210 N

32. The effect on an air gap in a magnetic circuit is to:

A. increase the reluctance
B. reduce the flux density
C. divide the flux
D. reduce the magnetomotive force

33. If a conductor is horizontal, the current flowing from left to right and the direction of the surrounding magnetic field is from above to below, the force exerted on the conductor is:

A. from left to right
B. from below to above
C. away from the viewer
D. towards the viewer

34. Electromotive force is provided by:

A. resistance's
B. a conducting path
C. an electric current
D. an electrical supply source

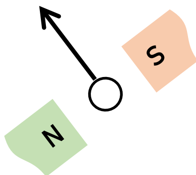
35. Find the speed that a conductor of length 120 mm must be moved at right angles to a magnetic field of flux density 0.6 T to induce in it an e.m.f. of 1.8 V.

A. 2.5 m/s
B. 0.25 m/s
C. 250 m/s
D. 25 m/s

36. The most basic particle of positive charge is the _____.

A. coulomb
B. proton
C. electron
D. neutron

37. For the current-carrying conductor lying in the magnetic field shown in the figure below, the direction of the current in the conductor is:



A. towards the viewer
B. away from the viewer
C. upwards
D. downwards

38. An electric bell depends for its action on:

A. a permanent magnet
B. reversal of current
C. a hammer and a gong
D. an electromagnet

39. A relay can be used to:

A. decrease a current in a circuit
B. control a circuit more readily
C. increase a current in a circuit
D. control a circuit in a distance

40. The magnetic field due to a current-carrying conductor takes form of:

A. rectangles
B. concentric circles
C. wavy lines
D. straight lines radiating outwards

41. A strong permanent magnet is plunged into a coil and left in the coil. What is the effect produced on the coil after a short time?

A. There is no effect
B. The insulation of the coil burns out
C. A high voltage is induced
D. The coil winding becomes hot

42. In what time would a current of 10 A transfer a charge of 50 C?

A. 5 s
B. 5 ms
C. 0.5 s
D. 0.5 ms

43. The atomic number of an element is determined by:

A. The number of neutrons.
B. The number of protons.
C. The number of neutrons plus the number of protons.
D. The number of electrons.

44. An ion:

A. Is electrically neutral.
B. Has positive electric charge.
C. Has negative electric charge.
D. Might have either a positive or negative charge.

45. There is a force of attraction between two current-carrying conductors when the current in them is:

A. in opposite direction
B. in the same direction
C. of different magnitude
D. of the same magnitude

46. A coulomb:

A. Represents a current of one ampere.
B. Flows through a 100-watt light bulb.
C. Is one ampere per second.
D. Is an extremely large number of charge carriers.

47. A potentially lethal electric current is on the order of _____.

A. 0.01 mA
B. 1 mA
C. 0.1 mA
D. 0.1 A

48. Which of the following units can represent magnetic flux density?

A. The volt-turn.
B. The ampere-turn.
C. The gauss.
D. The gauss-turn.

49. A ferromagnetic material:

A. Concentrates magnetic flux lines within itself.
B. Increases the total magnetomotive force around a current-carrying wire.
C. Causes an increase in the current in a wire.
D. Increases the number of ampere-turns in a wire.

50. The force between two electrically charged objects is called:

A. Electromagnetic deflection.
B. Electrostatic force.
C. Magnetic force.
D. Electroscopic force.

END